

P8-4 Calculative Foundation

PART - A Vector & matrix Fundamentals

1) Represent each Students Subjects Score as a vector?

ans

[Math, Science, English, History]

$U = [\text{Math, Science, English, History}]$

Student 1 % S001

$S001 = \{73, 66, 61, 62\}$

$US001 = [73, 66, 61, 62]$

$US002 = [71, 83, 95, 79]$

2 Compute

o Norm-1
S001

$U = [73, 66, 61, 62]$

$$\|V\|_1 = |73| + |66| + |61| + |62|$$

$$\|V\|_1 = 262$$

Norm-2

$$\|V\|_2 = \sqrt{73^2 + 66^2 + 61^2 + 62^2}$$

$$\Rightarrow \sqrt{5329 + 4356 + 3721 + 3844}$$

$$\Rightarrow \|V\|_2 = 131.34$$

o Dot Product and Angle between two Students Score vectors?

S_{001} and S_{002}

$$V = [73, 66, 61, 62]$$

$$W = [71, 83, 95, 79]$$

$$V \cdot W = V_1 W_1 + V_2 W_2 + V_3 W_3 + V_4 W_4$$

$$= (73)(71) + (66)(83) + (61)(95) + (62)(79)$$

$$= 5183 + 5478 + 5795 + 4898$$

$$\Rightarrow V \cdot W = 21354$$

Angle between Vectors

$$\cos(\theta) = \frac{V \cdot W}{\|V\|_2 \|W\|_2}$$

$$\theta \approx 9.63^\circ$$

So01 and So02 dono ke Score vectors ke beech angle approximately 9.63° hai. Seletively dono Similar direction mein hai,

9. Cross Product - 3D Selected Subjects

Cross Product me Sirf 3-dimensional Vectors ke Liye hota hai.

- MATH
- Science
- English

$$S001 \quad u = [73, 66, 61]$$

$$S002 \quad w = [71, 83, 95]$$

$$u \times w = \begin{bmatrix} i & j & k \\ 73 & 66 & 61 \\ 71 & 83 & 95 \end{bmatrix}$$

$$= [1207, -2604, 1373]$$

3 Find the Projection of one vector into another?

ans S_{001} , S_{002}

$$v = [73, 66, 61, 62]$$

$$w = [71, 83, 95, 79]$$

$$\text{Proj}_w (v) = \frac{v \cdot w}{w \cdot w} w$$

$$v \cdot w = 21354$$

$$\begin{aligned} w \cdot w &= 71^2 + 83^2 + 95^2 + 79^2 \\ &= 5041 + 6889 + 9025 + 6241 \\ &= 27196 \end{aligned}$$

$$\text{Proj}_w = \frac{21354}{27196} w$$

Coefficient approximately :-
 $\Rightarrow 0.7852$

$$P_{2016}(V) = [55.75, 65.17, 74.59, 62.03]$$

PART - B

Form a matrix of Students $\times$ Subjects
Perform

- Matrix addition and multiplication

200×4 matrix ko directly kisi Same Size 200×4 matrix ke Sath add karana Possible hai.

$A_1 = 1-100$ students

$A_2 = 101-200$ Students

$$A_1 = 100 \times 4$$

$$A_2 = 100 \times 4$$

$$[73, 66, 61, 62] + [82, 77, 87, 88]$$

$$= [155, 143, 148, 150]$$

matrix addition is performed element by element.

Teacher's Signature

matrix multiplication

$$A = 200 \times 4$$

$A \times A$ not possible because of inner dimensions.

$$4 \neq 200$$

We use transpose

$$A^T = 4 \times 200$$

$$\begin{aligned} A^T A \\ &= (4 \times 200)(200 \times 4) \\ &= (4 \times 4) \end{aligned}$$

o Transpose and inverse

$$A = 200 \times 4$$

$$\text{Transpose} = A^T = 4 \times 200$$

$$A = \begin{bmatrix} 73 & 66 & 61 & 62 \\ 71 & 83 & 95 & 79 \\ 69 & 78 & 69 & 66 \\ 87 & 85 & 80 & 74 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 73 & 71 & 69 & 87 \\ 66 & 83 & 78 & 85 \\ 61 & 95 & 69 & 80 \\ 62 & 79 & 66 & 74 \end{bmatrix}$$

$$A^T = 4 \times 200$$

o Inverse

$$A = 200 \times 4$$

square matrix ko form hota hai
 $n \times n$

but, $200 \neq 4$

Inverse of original 200×4 matrix
 is not possible.

Dem Determinant
4 Students $\times$ 4 Subjects Ka Square
matrix

$$A \begin{bmatrix} 73 & 66 & 61 & 62 \\ 71 & 83 & 95 & 79 \\ 69 & 78 & 69 & 66 \\ 87 & 85 & 80 & 74 \end{bmatrix}$$

$$\text{Determinant} = 102708$$

Since determinant $\neq 0$, inverse of this
4x4 matrix is possible.

PART C : Linear Transformation
Geometry.

Student $\rightarrow$ (MATH, Science, English, History)

$$S_{001, 2} (73, 66, 61, 62)$$

So, each Student represented as a 4D
Vector.

5 Explain Line, Plane, and hyperplane

1) Line

A Line is 1 dimensional

$$T(x) = x(73, 66, 61, 62)$$

This represents a line through the origin in the 4D Students Performance Space.

2) Plane

A Plane is 2 dimensional.

$$\begin{aligned} S_{001} &= [73, 66, 61, 62] \\ S_{002} &= [71, 83, 95, 79] \end{aligned}$$

Linear Combination

$$aS_{001} + bS_{002}$$

Form a 2D Plane inside the 4D data set Space.

3. Hyper Plane

A hyper Plane in 4D is 3 dimensional.

agar three independent Students ke vectors henge.

$$aS_{001} + bS_{002} + cS_{003}$$

creates a 3D hyper Plane / SubSpace inside the 4D Subject Space.

PART D: Eigen values & decomposition

1) Eigen values

Eigen values batata hai ki Covariance matrix mein kis direction mein data ka ~~var~~ variance / spread kitna hai.

Eigen Vector batata hai ki woh direction kya hai.

Covariance matrix
using all 200 students

$$C = \begin{bmatrix} 122.14 & 71.22 & 58.97 & 65.60 \\ 71.22 & 123.36 & 62.97 & 62.68 \\ 58.97 & 62.97 & 86.28 & 56.92 \\ 65.60 & 62.68 & 56.92 & 99.71 \end{bmatrix}$$

Eigen values

$$[299.149, 82.325, 45.722, 34.292]$$

Eigenvectors

Corresponding eigen vectors.

$$[-0.539, -0.543, -0.436, -0.473]$$

$$[-0.644, 0.729, 0.110, -0.205]$$

$$[0.542, 0.335, -0.338, -0.692]$$

$$[0.023, 0.247, -0.826, 0.506]$$

Conclusion : Sabse bada eigenvalue 299.149 hai, matlab dataset ki Sabse large variation first Principal direction mein hai.

The End.